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Inventor(s)

White; Chris

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### PAINTLESS DENT REMOVAL TOOL, SYSTEM AND METHOD

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#### Abstract

A paintless dent repair glue tab tool, system and method is disclosed. In general, the paintless dent repair glue tab tool is a hinge tab assembly having a first hinge tab, second hinge tab, hinge lift, and ball detention pin. The first hinge tab, second hinge tab and hinge lift each have rings connected thereto and for slidably engaging with and connecting to the ball detention pin. In use, the first hinge tab and the second hinge tab are positioned about a dent on the surface of a vehicle and are connected to the surface of the vehicle using glue. The hinge lift is then pulled upward by a separate paintless dent removal tool causing the first hinge tab and second hinge tab to rotate about the ball detention pin and pull up and out the dent on the surface of the vehicle.

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**Inventors:** White; Chris (Oklahoma City, OK)

**Applicant:** EHC INC. (Oklahoma City, OK)

**Family ID:** 1000008465230

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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Application No. 63/556,457, filed on Feb. 22, 2024, which is expressly incorporated herein by reference in its entirety.

### FIELD OF INVENTION

[0002] The invention is in the technical field of paintless dent repair tools, systems and methods. The invention pertains generally to paintless dent repair glue tab tools, systems and methods used to “pull-out” a dent from a vehicle body, such as an automobile, aircraft, motorcycle or the like, for example.

### BACKGROUND

[0003] Advancements in construction methods and materials have resulted in lighter and stronger vehicle bodies for automobiles, aircrafts, motorcycles and the like. However, such improved vehicle bodies are often increasingly prone to dents, dings and various other body deformations. Such damage may be caused by, for example, automobile accidents, hail or debris striking the outer surface of the vehicle body. Repairing such dents, dings and deformations can be difficult, laborious and time consuming, even when using traditional paintless dent repair tools and processes.

[0004] In general, paintless dent repair (sometimes known as “PDR”) is the process of repairing dents to a vehicle body without repainting the vehicle. Thus, saving the owner of the vehicle time and money by avoiding the cost and expense of repainting the vehicle. Traditionally, in paintless dent repair, many repair shops utilize the flexibility and durability of the outer surface of the vehicle body to reach into vehicle body compartments to “push-out” a dent from the inside outward or “pull-out” a dent from the outside of the vehicle body. Once the dent is pushed or pulled out, there is also a need to partially “knock-down” or flatten out the dent to completely remove the dent.

[0005] Traditionally, repair shops have used a variety of tools to help assist with the paintless dent repair process, including tools designed to assist a user with the “push-out” and “pull-out” process and tools designed to assist a user with the “knock-down” process. One such traditional tool is known in the art as a “glue tab.” Traditional glue tab tools are generally constructed from a single unitary piece of material and generally have a top member, a neck member and a flexible bottom member. Traditional glue tab tools are configured so that the flexible bottom member is affixed to a dent with glue and the top member is pulled upward by a user (either by hand or by machine) to pull-out a dent in the vehicle body. The neck member of a traditional glue tab tool is generally of a relatively large and uniform or standard size diameter (for ease of manufacturing and strength of the tool) regardless of the size of the dent to be removed and regardless of the size of the flexible bottom member. Traditionally, this is to prevent the neck member from snapping and breaking when the top member is pulled upward during use.

[0006] When a traditional glue tab tool pulls a glue tab that is affixed to a dent upward, the dent is pulled up around the circumference of the bottom of the neck member creating a curved arc or bump on the vehicle body where the dent is pulled-out. Typically, the arc or bump must then be flattened back down using a knock down tool and then pulled back out using another traditional glue tab tool, usually multiple times, before the dent is ultimately removed. It is common for such traditional glue tab tools to break or deform during use and glue tab tools are often constructed in a variety of different sizes and shapes for use with specific sized or shaped dents and specific sized or shaped tools. Thus, in use a variety of traditional glue tabs are typically used and each one is often used only once due to loss of structural integrity, deformation or breakage of the traditional glue tab. Thus, requiring multiple and different sized and shaped glue tabs and multiple and different sized and shaped tools to be used during the process. This process takes a significant

amount of time to complete, is tedious, cumbersome and labor intensive. Thereby, increasing the cost and decreasing the effectiveness and usefulness of such traditional glue tab tools.

[0007] To that end, it would be advantageous to provide an improved paintless dent repair glue tab tool, system and method that is configured to more efficiently and precisely pull-out dents and to improve the accuracy and reliability of the dent removal process. The improved paintless dent repair glue tab tool disclosed herein is an improved hinge tab assembly that includes a first and second hinge tab, a hinge lift, a ball detention pin, and in some embodiments a lateral tension tab block. The first hinge tab, second hinge tab, and hinge lift are detachably connected about the ball detention pin such that each of the first hinge tab, second hinge tab, and hinge lift are capable of rotating about the ball detention pin. The lateral tension tab block is detachably connected between the hinge lift and the first and second hinge tab to set and secure the hinge tab assembly at a desired angle and to provide structural support when the hinge lift is pulled upward by a paintless dent repair tool.

[0008] The first hinge tab, second hinge tab, and hinge lift each have a plurality of circular rings, sometimes referred to as knuckles or spirals positioned along a portion of the first hinge tab, second hinge tab and hinge lift. The rings are configured such that the rings on the first hinge tab, second hinge tab, and hinge lift align with one another, such that the ball detention pin may be selectively inserted through the rings of each of the first hinge tab, second hinge tab, and hinge lift to detachably connect the first hinge tab, second hinge tab, and hinge lift together and such that they are capable of rotating about the ball detention pin. The ball detention pin may be selectively inserted into the rings of each of the first hinge tab, second hinge tab and hinge lift and then locked into place by detachably inserting the lateral tension tab block between the hinge lift and the first and second hinge tab. The lateral tension tab block may include an opening for a screw, pin or bolt to be inserted therethrough and to tighten and secure the lateral tension tab block to the hinge tab assembly.

[0009] The hinge tab assembly and method of use described herein is configured to be reusable, modular, and selectively interchangeable with each of its component parts. Further, because each of the first hinge tab and second hinge tab rotate about the ball detention pin when the hinge lift is pulled upward, the hinge tab assembly has increased structural strength. Further the hinge tab assembly is reusable and each of the component parts of the hinge tab assembly may be selectively interchanged with other component parts of the hinge tab assembly in real time and on the fly by a user and for use with different sized and shaped dents. These advantages are unexpected and provide substantial benefits over traditional glue tabs which are not reusable, prone to breaking or deforming and which are not selectively modifiable or customizable for use with a variety of different sized or shaped dents. Moreover, traditional glue tabs are not formed with a ball detention pin that enables the portions of the glue tab that make contact with the vehicle to rotate about a centrally disposed ball detention pin to provide greater strength, durability, and reducing the likelihood of breakage, cracking or deformation of the tool. Thus, the hinge tab assembly and method of use provides numerous advantages and benefits over traditional glue tab tools. Thereby, significantly increasing the efficiency and reducing the cost of the paintless dent repair process.

#### SUMMARY OF THE INVENTION

[0010] The inventive concepts disclosed herein are generally directed to an improved paintless dent repair glue tab tool, system and method. In particular, a hinge tab assembly having a first and second hinge tab, a hinge lift, a ball detention pin, and in some embodiments a lateral tension tab block. The first hinge tab, second hinge tab, and hinge lift are detachably connected about the ball detention pin such that each of the first hinge tab, second hinge tab, and hinge lift are capable of rotating about the ball detention pin. The lateral tension tab block may be detachably connected between the hinge lift and the first hinge tab and the second hinge tab to set and secure the hinge tab assembly at a desired angle and to provide structural support to the hinge tab assembly when the hinge lift is pulled upward by a paintless dent repair tool. When the hinge tab assembly is

pulled upward by a paintless dent removal tool, the first hinge tab and the second hinge tab, which are detachably connected to a vehicle by glue, are pulled upward, lifting the dent out of the vehicle. Because the hinge tab assembly includes a ball detention pin which detachably connects the first hinge tab and the second hinge tab to the hinge lift, the hinge tab assembly is capable of rotating about the ball detention pin without breaking, deforming or cracking the hinge tab assembly. Thus, permitting the hinge tab assembly to pull up and out dents using greater force and without breaking, deforming or cracking the connection points between the first hinge tab, second hinge tab, and hinge lift. Thereby providing significant improvements over traditional glue tab tools.

[0011] The first hinge tab, second hinge tab, and hinge lift each have a plurality of circular rings, sometimes referred to as knuckles or spirals positioned along a portion of the first hinge tab, second hinge tab and hinge lift. The rings are configured such that the rings on the first hinge tab, second hinge tab, and hinge lift align with one another, such that the ball detention pin may be selectively inserted through the rings of each of the first hinge tab, second hinge tab, and hinge lift to detachably connect the first hinge tab, second hinge tab, and hinge lift together and such that they are capable of rotating about the ball detention pin. The ball detention pin may be selectively inserted into the rings of each of the first hinge tab, second hinge tab and hinge lift and then locked into place by detachably inserting the lateral tension tab block between the hinge lift and the first and second hinge tab. The lateral tension tab block may include an opening for a screw, pin or bolt to be inserted therethrough and to tighten and secure the lateral tension tab block to the hinge tab assembly.

[0012] A method of connecting the various component parts of the hinge tab assembly and method of use are also disclosed. In use, the desired first hinge tab, second hinge tab, hinge lift, ball detention pin and lateral tension tab block are selected from a plurality of different sized, shaped, and configured, first hinge tab, second hinge tab, hinge lift, ball detention pin and lateral tension tab block and are detachably connected to one another to form a specific and unique hinge tab assembly, which has been selectively assembled by a user based on the size, shape and location of the dent to be removed. The size, shape, and configuration of each of the first hinge tab, second hinge tab, hinge lift, ball detention pin, and lateral tension block that is selected by a user is based upon the size and shape of the dent to be removed and the position of said dent on the vehicle. The first hinge tab and the second hinge tab are detachably secured on the vehicle using glue such that when the hinge lift is pulled upward by a paintless dent removal tool, the dent will be substantially removed. A variety of different paintless dent removal tools can be used to detachably connect to the hinge lift and pull the hinge lift upward, pulling out and substantially removing the dent of the surface of the vehicle. Thus, making the hinge tab assembly a versatile and efficient tool.

[0013] When the hinge tab assembly is pulled upward by the hinge lift, the ball detention pin positioned at the bottom of the hinge lift, pulls each of the first and second hinge tab upward, creating an upside down “v” or “Y” shape without breaking, cracking, or bending the hinge tab assembly because each of the first hinge tab and the second hinge tab are able to rotate about the ball detention pin when the hinge lift is pulled upward. Traditional glue tab tools are not capable of rotating about a connection point between the portion that is detachably connected to the vehicle and the portion that lifts the glue tab upward, primarily because such traditional glue tab tools are formed from a single unitary piece of material. Thus, traditional glue tab tools are prone to breaking, cracking and deforming during use due to the pressure applied to the glue tab tool when pulled upward and such traditional glue tab tools are generally not able to be reused because the structural integrity of the tool becomes damaged during the dent removal process. Thus, such traditional dent removal tools are cumbersome and significantly less effective at removing dents.

[0014] Because the first hinge tab and the second hinge tab are detachably connected about the ball detention pin and are capable of rotating about the ball detention pin, without breaking or bending, the hinge tab assembly has significant and substantial advantages over traditional glue tab tools. For example, the disclosed hinge tab assembly has significantly increased structural strength and a

longer life span of use because the component parts of the hinge tab assembly may be easily and efficiently reused, replaced, disassembled, and reassembled without breaking, deforming or loss of structural strength.

[0015] Further, traditional glue tab tools are typically formed or constructed from a single unitary piece of material, which causes the glue tab to break, deform, and crack when pulled upward. Further, traditional glue tabs are constructed to be a specific size and shape and are therefore only usable only with a specific size, shape or location of dent. Thus, traditional glue tab tools are not selectively customizable, interchangeable, reusable in the way that the hinge tab assembly disclosed herein is. The hinge tab assembly provides significant cost, efficiency, and durability advantages over existing traditional glue tab tools. Because traditional glue tab tools are typically constructed from a single unitary piece of material such glue tab tools are prone to breaking, deforming or cracking at the connection points between the portion that is detachably glued to the vehicle and the portion that is pulled upward by a dent removal tool. In contrast, the hinge tab assembly disclosed herein is capable of rotating about the ball detention pin to provide significantly enhanced structural strength and durability during use without breaking, deforming, or cracking the hinge tab assembly. Further, because each of the component parts of the hinge tab assembly may be selectively detachably connected, each of the component parts of the hinge tab assembly are removable, replaceable, and capable of being selectively modified, interchanged, and reused with a variety of different sized and shaped component parts. Thus, providing maximum durability and versatility because a user can adjust, modify, configure and select a variety of different sized and shaped component parts for use in the hinge tab assembly on the fly and in real time and then may reuse such component parts multiple times in the same or different configurations for a variety of different sized and shaped dents.

[0016] It should be understood that each of the component parts of the hinge tab assembly, namely the first hinge tab, second hinge tab, hinge lift, ball detention pin, and lateral tension tab block are configured to be selectively and detachably connected, disconnected, and used interchangeably with a variety of different sized and shaped hinge tab, hinge lift, ball detention pin, and lateral tension tab block to create a customizable, modular, and selectively interchangeable system.

[0017] In this way, a user may, depending upon the size, shape and position of a particular dent to be removed, select from a range of differently configured, sized, and shaped hinge tab, hinge lift, ball detention pin, and lateral tension tab block, on the fly and in real time, to detachably connect and assembly a specific hinge tab assembly configured for use with a particular sized and shaped dent. In the event any component part of the hinge tab assembly breaks, or a user desires to remove a different sized or shape dent, a user may selectively remove any component part and replace it with another component part or customize a new configuration of the hinge tab assembly in real time, easily and efficiently without having to use a completely different tool. Thus, numerous configurations of the hinge tab assembly are contemplated and may be selectively arranged and customized by a user for maximum flexibility and efficiency. Further, because the size, shape and configuration of the hinge tab assembly may be selectively customized by a user, a variety of different sized and shaped paintless dent removal tools may be used in conjunction with the hinge tab assembly to pull out dents in the vehicle.

[0018] In some embodiments, the hinge tab assembly, including the hinge lift, first hinge tab and second hinge tab may also include bracing rods, members or filaments positioned within or upon the hinge lift, first hinge tab and second hinge tab. The bracing rods, members or filaments are configured to provide additional structural strength to the hinge assembly during use.

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## Description

## BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0019] Like reference numerals in the figures represent and refer to the same or similar element or function. Implementations of the disclosure may be better understood when consideration is given to the following detailed description thereof. Such description makes reference to the annexed pictorial illustrations, schematics, graphs, drawings, and appendices. In the drawings:

[0020] FIG. 1 is an exploded view of an embodiment of a paintless dent repair glue tab tool (100) in accordance with the inventive concepts disclosed herein.

[0021] FIG. 2A is a perspective view of an embodiment of a paintless dent repair glue tab tool (100) in accordance with the inventive concepts disclosed herein.

[0022] FIG. 2B is a perspective view of the paintless dent repair glue tab tool (100) of FIG. 2A in accordance with the inventive concepts disclosed herein.

[0023] FIG. 3A is a perspective view of a hinge lift (135) of an embodiment of a paintless dent repair glue tab tool (100) in accordance with the inventive concepts disclosed herein.

[0024] FIG. 3B is a perspective view of the hinge lift (135) of FIG. 3A in accordance with the inventive concepts disclosed herein.

[0025] FIG. 4A is a perspective view of a second hinge tab (125) of an embodiment of a paintless dent repair glue tab tool (100) in accordance with the inventive concepts disclosed herein.

[0026] FIG. 4B is a perspective view of the second hinge tab (125) of FIG. 4A in accordance with the inventive concepts disclosed herein.

[0027] FIG. 4C is a perspective view of the second hinge tab (125) of FIG. 4A in accordance with the inventive concepts disclosed herein.

[0028] FIG. 5 is a perspective view of an embodiment of a paintless dent repair glue tab tool (100) detachably connected to a lift mechanism (99) wherein the angle of the first hinge tab (115) and second hinge tab (125) of the hinge tab assembly (110) is about 281°, in accordance with the inventive concepts disclosed herein.

[0029] FIG. 6 is a perspective view of an embodiment of a paintless dent repair glue tab tool (100) detachably connected to a lift mechanism (99) wherein the angle of the first hinge tab (115) and second hinge tab (125) of the hinge tab assembly (110) is about 90°, in accordance with the inventive concepts disclosed herein.

[0030] FIG. 7 is a perspective view of a method (300) of assembling a paintless dent repair glue tab tool (100) in accordance with the inventive concepts disclosed herein.

[0031] FIG. 8 is a perspective view of a method (400) of removing a dent from a vehicle in accordance with the inventive concepts disclosed herein.

## DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0032] Before explaining at least one embodiment of the inventive concepts disclosed herein in detail, it is to be understood that the inventive concepts are not limited in their application to the details of construction and the arrangements of the components or steps or methodologies set forth in the following description or illustrated in the drawings. The inventive concepts disclosed herein are capable of other embodiments or of being practiced or carried out in various ways. Also, it is to be understood that the phraseology and terminology employed herein is for the purpose of description and should not be regarded as limiting the inventive concepts claimed herein in any way.

[0033] In the following detailed description of embodiments of the inventive concepts, numerous specific details are set forth in order to provide a more thorough understanding of the inventive concepts. However, it will be apparent to one of ordinary skill in the art that the inventive concepts within the disclosure may be practiced without these specific details. In other instances, well-known features have not been described in detail to avoid unnecessarily complicating the instant disclosure.

[0034] As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,”

“having,” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed.

[0035] Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by anyone of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

[0036] In addition, use of the “a” or “an” are employed to describe elements and components of the embodiments herein. This is done merely for convenience and to give a general sense of the inventive concepts. This description should be read to include one or at least one and the singular also includes the plural unless it is obvious that it is meant otherwise.

[0037] Finally, as used herein any reference to “one embodiment” or “an embodiment” means that a particular element, feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment.

[0038] Referring now to FIGS. 1-8, shown therein is a perspective view of an embodiment of a paintless dent repair glue tab tool (**100**) for removing dents from a vehicle. The paintless dent repair glue tab tool (**100**) is a hinge tab assembly (**110**). The hinge tab assembly (**110**) includes a first hinge tab (**115**). The first hinge tab (**115**) has a first end (**116**), a second end (**117**), a first side (**118**), a second side (**119**), a top surface (**120**), a bottom surface (**121**), and a plurality of rings (**122**) positioned along the first side (**118**) of the first hinge tab (**115**). The plurality rings (**122**) are configured for receiving a ball detention pin (**123**) therethrough.

[0039] The hinge tab assembly (**110**) further includes a second hinge tab (**125**). The second hinge tab (**125**) has a first end (**126**), a second end (**127**), a first side (**128**), a second side (**129**), a top surface (**130**), a bottom surface (**131**), and a plurality of rings (**132**) positioned along the first side (**128**) of the second hinge tab (**125**). The plurality of rings (**132**) are configured for receiving a ball detention pin (**123**) therethrough.

[0040] The hinge tab assembly (**110**) further includes a hinge lift (**135**). The hinge lift (**135**) has a first end (**136**), a second end (**137**), a first side (**138**), a second side (**139**), a top surface (**140**), a bottom surface (**141**), and a plurality of rings (**142**) positioned along the bottom surface (**141**). The hinge lift (**135**) is configured for receiving a ball detention pin (**123**) therethrough.

[0041] The hinge tab assembly (**110**) further includes a ball detention pin (**123**). The ball detention pin (**123**) is detachably connected to the first hinge tab (**115**), the second hinge tab (**125**), and the hinge lift (**135**), by insertion of the ball detention pin (**123**) through the plurality of rings (**122**) on the first hinge tab (**115**), the plurality of rings (**132**) of the second hinge tab (**125**), and the plurality of rings (**142**) of the hinge lift (**135**). Wherein, the first hinge tab (**115**), second hinge tab (**125**), and hinge lift (**135**) rotate about the ball detention pin (**123**). Wherein, the bottom surface (**121**) of the first hinge tab (**115**) and the bottom surface (**131**) of the second hinge tab (**125**) are configured to be detachably connected to a vehicle by glue. The hinge lift (**135**) is configured to be detachably connected to a paintless dent removal tool and pulled upward, wherein the first hinge tab (**115**) and the second hinge tab (**125**) rotate about the ball detention pin (**123**) to pull up and out a dent on a vehicle at a peak and without breaking or deforming the hinge tab assembly (**110**).

[0042] In some embodiments, the hinge tab assembly (**110**) further includes a lateral tension tab block (**150**) detachably connected to the hinge tab assembly (**110**). The lateral tension tab block (**150**) has a first end (**152**), a second end (**154**), a first side (**156**), a second side (**158**), a top surface (**160**) and a bottom surface (**162**). The lateral tension tab block (**150**) has a plurality of recesses (**164**) positioned on the first side (**156**) of the lateral tension tab block (**150**) and the plurality of recesses (**164**) are oriented for receiving and detachably connecting to the plurality of rings (**142**) of the hinge lift (**135**). Wherein the lateral tension tab block (**150**) is configured to provide

structural rigidity to the hinge tab assembly (110) and substantially prevent the hinge lift (135) from rotating about the ball detention pin (123) during use.

[0043] In some embodiments of the hinge tab assembly (110), the first hinge tab (115) and the second hinge tab (125) are the same size; while in other embodiments of the hinge tab assembly (110), the first hinge tab (115) and the second hinge tab (125) are different sizes.

[0044] The hinge tab assembly (110), including the first hinge tab (115), the second hinge tab (125) and the hinge lift (135), and ball detention pin (123) may be constructed from any material of sufficient strength necessary to maintain the structural integrity of the hinge tab assembly (110) and which is sufficiently flexible to permit the hinge tab assembly (110) to rotate about the ball detention pin (123) and to bend and flex without breaking or deforming during use. For example, the first hinge tab (115), second hinge tab (125) and hinge lift (135) may be constructed from non-metals such as plastics, polycarbonate, impact resistant nylon, carbon fiber, polymers, resins, ceramics, composite materials, or metals such as alloys, steel, titanium, combinations thereof and the like. Further, the ball detention pin (123) may be constructed from metals such as alloys, steel, or titanium and/or non-metals such as plastics, polycarbonate, impact resistant nylon, carbon fiber, polymers, resins, ceramics, composite materials, combinations thereof and the like.

[0045] The hinge tab assembly (110) is shown as having a substantially rectangular shape.

However, it should be understood that the hinge tab assembly (110) may have any desired shape, including but not limited to, a generally rectangular, square, circular, or triangular shape for example. Further, the first hinge tab (115), the second hinge tab (125) and the hinge lift (135) may each have the same or a different shape and size from one another. For example, the first hinge tab (115) may be substantially rectangular in shape while the second hinge tab (125) may be substantially square in shape while the hinge lift (135) may be substantially triangular in shape for example. Further, the dimensions and size of each of the first hinge tab (115), second hinge tab (125), and hinge lift (135) may be the same or different dimensions, such that the width, length, thickness, height, may be the same or different dimension from each other.

[0046] It is contemplated that a user will select from a variety of different sized, shaped and dimensioned, first hinge tab (115), second hinge tab (125) and hinge lift (135) to selectively create a customized hinge tab assembly (110) in real time and on the fly based upon the size, shape and location of the dent to be removed. In this way, different sized and shaped configurations of the hinge tab assembly (110) may be constructed on the fly and in real time by a user selectively choosing from a plurality of different sized and shaped first hinge tab (115), second hinge tab (125), hinge lift (135) and ball detention pin (123) to create a custom paintless dent repair glue tab tool (100) specific for a particular sized, shaped, and positioned dent on a vehicle.

[0047] In some embodiments, the hinge tab assembly (110) further includes additional bracing, ribs, or other structural support for increased strength. For example, the first hinge tab (115), second hinge tab (125) and hinge lift (135) may each include additional reinforcement, either integrated into the structure or formed thereon, including but not limited to, ridges, ribs, bracing structures, pins, layered structural component within the structure, or other components configured to increase the strength, durability or flexibility of the hinge tab assembly (110) and to prevent deformation, cracking, or breakage of the hinge tab assembly (110) during use and when the hinge lift (135) is pulled upward by a paintless dent removal tool, including by a hand-held, machine operated, or other fully or partially automated paintless dent repair tool.

[0048] Further, in some embodiments the ball detention pin (123) may be formed from a variety of different materials to provide different levels of strength. In some embodiments, the ball detention pin (123) may also include one or more locking mechanisms or structures, such as pins, bolts and the like, to lock the ball detention pin (123) into place on the hinge tab assembly (110) or to permit a limited or specific range or angle of rotation of the first hinge tab (115), second hinge tab (125), and hinge lift (135) around the ball detention pin (123). Further, in some embodiments the ball detention pin (123) may be circular, square, rectangular or triangular so as to provide limited range



or angle of motion of the first hinge tab (115), second hinge tab (125), and hinge lift (135) about the ball detention pin (123).

[0049] In some embodiments, the dimensions of the first hinge tab (115), second hinge tab (125) and hinge lift (135) of the hinge tab assembly (110) are selected from a plurality of first hinge tab (115), second hinge tab (125) and hinge lift (135) having the following dimensions: 146×19 mm, 146×13 mm, 146×7 mm, 146×10 mm, 70×19mm, 70×13 mm, 70×7 mm, 25×25 mm, 25×19 mm, 25×13 mm, and 25×7 mm, for example. It should be understood that the dimensions of each of the first hinge tab (115), second hinge tab (125) and hinge lift (135) may be varied and are interchangeable so that the first hinge tab (115), second hinge tab (125) and hinge lift (135) may be the same dimensions or may be different dimensions.

[0050] Further, it should be understood that the plurality of rings (122) on the first hinge tab (115), the plurality of rings (132) on the second hinge tab (125) and the plurality of rings (142) on the hinge lift (135) may be of a variety of different sizes, shapes, and dimensions so as to fit a variety of different sized and shaped ball detention pin (123). Further, the number of rings for each of the first hinge tab (115), second hinge tab (125), and hinge lift (135) may be varied based on the dimensions of the first hinge tab (115), second hinge tab (125), and hinge lift (135). For example, in some embodiments, the first hinge tab (115), second hinge tab (125) and hinge lift (135) may contain one ring, three rings, four rings, six rings or twelve rings, for example. The number of rings may be varied and optimized based upon the desired size of the dent to be removed and a user may select from a plurality of first hinge tab (115), second hinge tab (125) and hinge lift (135) that contain a variety of different number of rings and different sized rings for use with a different sized ball detention pin(s) (123).

[0051] Further, it should be understood that the angle that the first hinge tab (115) and second hinge tab (125) rotate about the ball detention pin (123) may vary. In some embodiments, the angle of the first hinge tab (115) and second hinge tab (125) may rotate is 281° degrees; while in other embodiments the angle of the first hinge tab (115) and second hinge tab (125) may rotate is 90° degrees. The angle of rotation about the ball detention pin (123) is limited by size and position of the hinge lift (135), the size of the first hinge tab (115), second hinge tab (125), and the size of the rings positioned on each of the first hinge tab (115), second hinge tab (125) and hinge lift (135). Further, the angle of rotation may be selectively set by a user to a specific angle of rotation by using a locking mechanism or pin.

[0052] Further, in some embodiments, the hinge tab assembly (100) also includes a lateral tension tab block (150). The lateral tension tab block (150) detachably connected to the hinge tab assembly (110). The lateral tension tab block (150) having a first end (152), a second end (154), a first side (156), a second side (158), a top surface (160) and a bottom surface (162). The lateral tension tab block (150) having a plurality of recesses (164) positioned on the first side (156) of the lateral tension tab block (150) and oriented for receiving and detachably connecting of the plurality of rings (142) of the hinge lift (135). The lateral tension tab block (150) is configured to provide structural rigidity to the hinge tab assembly (110) and substantially prevent the hinge lift (135) from rotating about the ball detention pin (123) during use.

[0053] The lateral tension tab block (150) may be constructed from any material of sufficient strength necessary to prevent the first hinge tab (115) and second hinge tab (125) from rotating about the ball detention pin (123) or to limit the angle of rotation about the ball detention pin (123) and without breaking, cracking, or deforming the hinge tab assembly (110). For example, the lateral tension tab block may be constructed from non-metals such as plastics, polycarbonate, impact resistant nylon, carbon fiber, polymers, resins, ceramics, composite materials, or metals such as alloys, steel, titanium, combinations thereof and the like.

[0054] The lateral tension tab block (150) is shown as having a substantially rectangular shape wherein the plurality of recesses (164) are configured to align with the plurality of rings on the hinge lift (135). However, it should be understood that the lateral tension tab block (150) may have

any desired shape, including but not limited to, a generally rectangular, square, circular, or triangular shape for example. Further, the lateral tension tab block (150) may have the same or a different shape from the first hinge tab (115), the second hinge tab (125) and the hinge lift (135). [0055] It is contemplated that a user will select from a variety of different sized, shaped and dimensioned lateral tension tab block (150), to selectively create a customized hinge tab assembly (110) in real time and on the fly based upon the size, shape and location of the dent to be removed. In this way, different sized and shaped configurations of the hinge tab assembly (110) may be constructed on the fly and in real time by a user selectively choosing from a plurality of different sized and shaped first hinge tab (115), second hinge tab (125), hinge lift (135), ball detention pin (123) and lateral tension tab block (150) to create a custom paintless dent repair glue tab tool (100) specific for a particular sized, shaped, and positioned dent on a vehicle.

[0056] In use, the hinge tab assembly (110) is configured to be detachably connected to a paintless dent removal tool. In some embodiments, the hinge lift (135) may slidably engage with a lift mechanism (99), to detachably connected the hinge tab assembly (110) to a paintless dent removal tool, such as a hand-held paintless dent removal tool, an automated paintless dent removal tool, and/or a partially or fully automated paintless dent removal tool.

[0057] Referring now to FIGS. 1-10, shown therein is a method (300) of assembling a paintless dent repair glue tab tool (100). The method (300) includes the steps of: (A) selecting (302), from a plurality of different sized and shaped hinge tabs, a first hinge tab (115) and a second hinge tab (125). The first hinge tab (115) having a first end (116), a second end (117), a first side (118), a second side (119), a top surface (120), a bottom surface (121), and a plurality of rings (122) positioned along the first side (118). The plurality of rings (122) configured for receiving a ball detention pin (123) therethrough. The second hinge tab (125) has a first end (126), a second end (127), a first side (128), a second side (129), a top surface (130), a bottom surface (131), and a plurality of rings (132) positioned along the first side (128) of the second hinge tab (125). The plurality of rings (132) configured for receiving a ball detention pin (123) therethrough. The method (300) further includes the step of: (B) selecting (304), from a plurality of different sized and shaped hinge lifts, a hinge lift (135) having a first end (136), a second end (137), a first side (138), a second side (139), a top surface (140), a bottom surface (141), and a plurality of rings (142) positioned along the bottom surface (141). The plurality of rings (142) configured for receiving a ball detention pin (123) therethrough. The method (300) further including the step of: (C) connecting (306) the first hinge tab (115), the second hinge tab (125), and the hinge lift (135) to each other to form a hinge tab assembly (110). The first hinge tab (115), second hinge tab (125) and hinge lift (135) are detachably connected to one another by insertion of and slidably engaging a ball detention pin (123) through the plurality of rings (122) of the first hinge tab (115), the plurality of rings (132) of the second hinge tab (125), and the plurality of rings (142) of the hinge lift (135), such that the first hinge tab (115), second hinge tab (125), and hinge lift (135) are capable of rotating about the ball detention pin (123).

[0058] Referring now to FIGS. 1-8, shown therein is a method (400) of removing a dent from a vehicle. The method (400) including the steps of: (A) selecting (405), from a plurality of different sized and shaped hinge tabs, a first hinge tab (115) and a second hinge tab (125). The first hinge tab (115) having a first end (116), a second end (117), a first side (118), a second side (119), a top surface (120), a bottom surface (121), and a plurality of rings (122) positioned along the first side (118). The plurality of rings (122) configured for receiving a ball detention pin (123) therethrough. The second hinge tab (125) having a first end (126), a second end (127), a first side (128), a second side (129), a top surface (130), a bottom surface (131), and a plurality of rings (132) positioned along the first side (128) of the second hinge tab (125). The plurality of rings (132) configured for receiving a ball detention pin (123) therethrough. The method (400) further including the step of: (B) selecting (410), from a plurality of different sized and shaped hinge lifts, a hinge lift (135). The hinge lift (135) having a first end (136), a second end (137), a first side (138), a second side (139),

a top surface (140), a bottom surface (141), and a plurality of rings (142) positioned along the bottom surface (141). The plurality of rings (142) configured for receiving a ball detention pin (123) therethrough. The method (400) further including the step of: (C) connecting (415) the first hinge tab (115), the second hinge tab (125), and the hinge lift (135) to each other to form a hinge tab assembly (110).

[0059] The first hinge tab (115), second hinge tab (125) and hinge lift (135) are detachably connected to one another by insertion of and slidably engaging a ball detention pin (123) through the plurality of rings (122) of the first hinge tab (115), the plurality of rings (132) of the second hinge tab (125), and the plurality of rings (142) of the hinge lift (135), such that the first hinge tab (115), second hinge tab (125), and hinge lift (135) are capable of rotating about the ball detention pin (123). The method (400) further including the step of: (D) connecting (420) the first hinge tab (115) and the second hinge tab (125) to the surface of a vehicle with glue. The method (400) further includes the step of: (E) pulling (425) the hinge lift (135) upward, using a glue tab tool, to substantially remove a dent from the surface of the vehicle.

[0060] In use, and when looking at the hinge tab assembly from a side perspective view, when the hinge tab assembly (110) is pulled upward by the hinge lift (135), the ball detention pin (123) positioned at the bottom of the hinge lift, pulls each of the first hinge tab (115) and second hinge tab (125) upward, creating an upside down “v” or upside down “Y” shape. The first hinge tab (115) and the second hinge tab (125) are detachably connected to a vehicle with glue and when the hinge lift (135) is pulled upward, the hinge tab assembly (110) is able to pull upward with more force without breaking, cracking or bending the hinge tab assembly (110). This is because the first hinge tab (115) and the second hinge tab (125) are detachably connected about the ball detention pin (123) and are capable of rotating about the ball detention pin (123), instead of breaking, cracking, or bending thereby pulling up and out the dent on the surface of the vehicle more effectively and efficiently and permitting the various component parts of the hinge tab assembly (110) to be reused and interchanged. The hinge tab assembly (110) has significantly improved strength over traditional glue tab tools which are formed from an integral one piece construction and/or which are rigidly connected and thus not capable of rotating about a center point, such as the ball detention pin (123), for improved strength and durability. For example, the disclosed hinge tab assembly has significantly increased structural strength and a longer life span of use because the component parts of the hinge tab assembly may be easily and efficiently reused, replaced, disassembled, and reassembled without breaking, deforming or loss of structural strength.

[0061] It is to be appreciated that embodiments of the inventive concepts disclosed herein may be shipped fully or partially assembled, or fully or partially disassembled in the form of a kit, as will be readily appreciated by persons of ordinary skill in the art having the benefit of the instant disclosure.

[0062] From the above description, it is clear that the inventive concepts disclosed herein are adapted to carry out the objects and to attain the advantages mentioned herein as well as those inherent in the inventive concepts disclosed herein. While exemplary embodiments of the inventive concepts disclosed herein have been described for purposes of this disclosure, it will be understood that numerous changes may be made which will readily suggest themselves to those skilled in the art and which are accomplished within the broad scope of the inventive concepts disclosed herein and defined by the appended claims.

## Claims

1. A paintless dent repair glue tab tool for removing dents from a vehicle, the paintless dent repair glue tab tool comprising: a hinge tab assembly comprising: a first hinge tab, the first hinge tab having a first end, a second end, a first side, a second side, a top surface, a bottom surface, and a plurality of rings positioned along the first side of the first hinge tab, the plurality rings configured

for receiving a ball detention pin therethrough; a second hinge tab, the second hinge tab having a first end, a second end, a first side, a second side, a top surface, a bottom surface, and a plurality of rings positioned along the first side of the second hinge tab, the plurality of rings configured for receiving a ball detention pin therethrough; a hinge lift, the hinge lift having a first end, a second end, a first side, a second side, a top surface, a bottom surface, and a plurality of rings positioned along the bottom surface, the plurality of rings configured for receiving a ball detention pin therethrough; a ball detention pin, the ball detention pin detachably connected to the first hinge tab, the second hinge tab, and the hinge lift, by insertion of the ball detention pin through the plurality of rings on the first hinge tab, second hinge tab, and hinge lift, wherein, the first hinge tab, second hinge tab, and hinge lift rotate about the ball detention pin; and wherein, the bottom surface of the first hinge tab and the bottom surface of the second hinge tab are configured to be detachably connected to a vehicle by glue and the hinge lift is configured to be detachably connected to a paintless dent removal tool and pulled upward, wherein the first hinge tab and the second hinge tab rotate about the ball detention pin to pull up and out a dent on a vehicle at a peak.

2. The hinge tab assembly of claim 1 further comprising: a lateral tension tab block detachably connected to the hinge tab assembly, the lateral tension tab block having a first end, a second end, a first side, a second side, a top surface and a bottom surface, the lateral tension tab block having a plurality of recesses positioned on the first side of the lateral tension tab block and oriented for receiving and detachably connecting to the plurality of rings of the hinge lift; and wherein the lateral tension tab block is configured to provide structural rigidity to the hinge tab assembly and substantially prevent the hinge lift from rotating about the ball detention pin during use.

3. The hinge tab assembly of claim 1 wherein the first hinge tab and the second hinge tab are the same size.

4. The hinge tab assembly of claim 1 wherein the first hinge tab and the second hinge tab are different sizes.

5. A method of assembling a paintless dent repair glue tab tool, the method comprising the steps of: selecting, from a plurality of different sized and shaped hinge tabs, a first hinge tab and a second hinge tab, the first hinge tab having a first end, a second end, a first side, a second side, a top surface, a bottom surface, and a plurality of rings positioned along the first side, the plurality of rings configured for receiving a ball detention pin therethrough and a second hinge tab, the second hinge tab having a first end, a second end, a first side, a second side, a top surface, a bottom surface, and a plurality of rings positioned along the first side of the second hinge tab, the plurality of rings configured for receiving a ball detention pin therethrough; selecting, from a plurality of different sized and shaped hinge lifts, a hinge lift having a first end, a second end, a first side, a second side, a top surface, a bottom surface, and a plurality of rings positioned along the bottom surface, the plurality of rings configured for receiving a ball detention pin therethrough; and connecting the first hinge tab, the second hinge tab, and the hinge lift to each other to form a hinge tab assembly, by slidably engaging a ball detention pin through the plurality of rings of the first hinge tab, the plurality of rings of the second hinge tab, and the plurality of rings of the hinge lift, such that the first hinge tab, second hinge tab, and hinge lift are capable of rotating about the ball detention pin.

6. A method of removing a dent from a vehicle, the method comprising the steps of: selecting, from a plurality of different sized and shaped hinge tabs, a first hinge tab and a second hinge tab, the first hinge tab having a first end, a second end, a first side, a second side, a top surface, a bottom surface, and a plurality of rings positioned along the first side, the plurality of rings configured for receiving a ball detention pin therethrough and the second hinge tab having a first end, a second end, a first side, a second side, a top surface, a bottom surface, and a plurality of rings positioned along the first side of the second hinge tab, the plurality of rings configured for receiving a ball detention pin therethrough; selecting, from a plurality of different sized and shaped hinge lifts, a hinge lift having a first end, a second end, a first side, a second side, a top surface, a bottom

surface, and a plurality of rings positioned along the bottom surface, the plurality of rings configured for receiving a ball detention pin therethrough; connecting the first hinge tab, the second hinge tab, and the hinge lift to each other to form a hinge tab assembly, by slidably engaging a ball detention pin through the plurality of rings of the first hinge tab, the plurality of rings of the second hinge tab, and the plurality of rings of the hinge lift, such that the first hinge tab, second hinge tab, and hinge lift are capable of rotating about the ball detention pin; connecting the first hinge tab and the second hinge tab to the surface of a vehicle with glue; and pulling the hinge lift upward, using a glue tab tool, to substantially remove a dent from the surface of the vehicle.

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